METIS actor for MUSCLE 3

Introduction

This document describes how to use the actor interfacing METIS and MUSCLE3 (https://muscle3.readthedocs.io/en/latest/). It is part of METIS documentation localized in the directory “doc” under the root directory of METIS installation.

The METIS actor for MUSCLE 3 exchange messages with other actors formatted as IDSs following the standard defined in IMAS (https://github.com/iterorganization) .

Actor overview

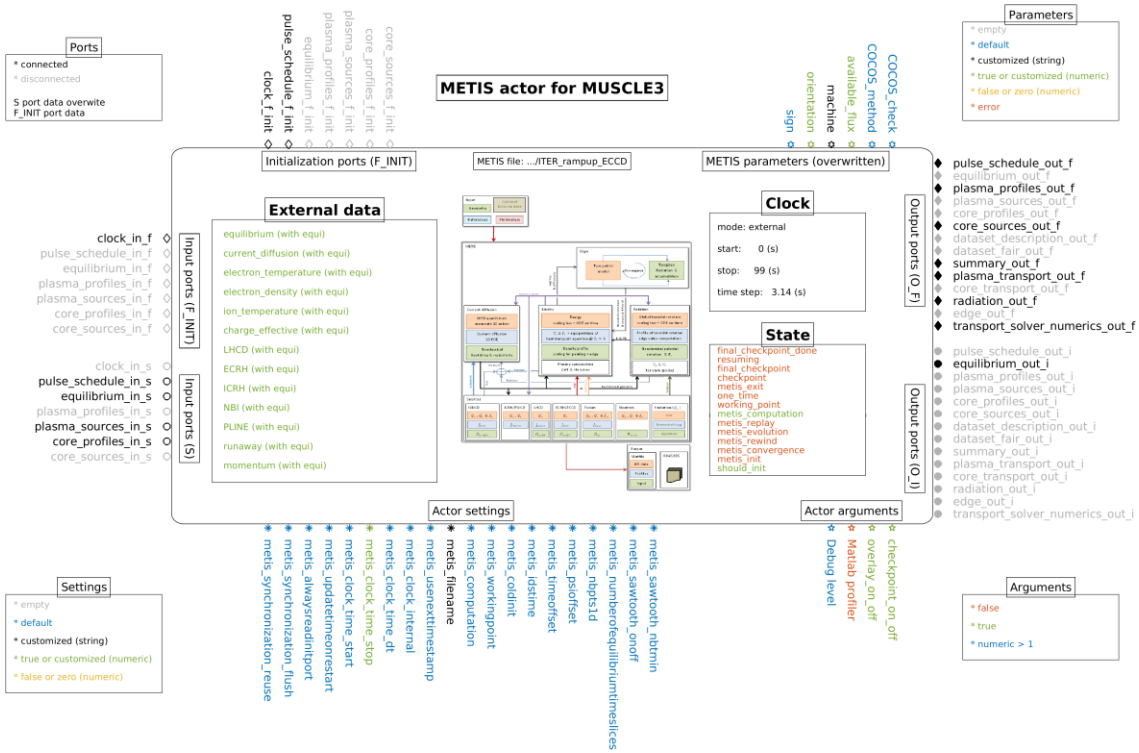


Figure 1: Layout of METIS actor for MUSCLE 3: this figure is dynamically generated at each run of the METIS actor and stored in the working directory of the METIS actor instance and is named “METIS_actor_for_MUSCLE3_layout.png”

The actor implements the following computation modes:

- resume
  Restart from a previously stopped simulation.

- replay
  Replays a METIS simulation file and broadcast IDSs messages preserving the time_stamp from the original simulation.

- interpolate
  Broadcast IDSs messages from a METIS simulation file interpolating on the time stamp

provided by MUSCLE3 workflow.

- working_point
  Compute the working point solution (the stationary time independent solution). The main purpose of this mode is the coupling METIS to a system code.

- metis_init
  Initialization of the METIS actor. There two ways to do it:
  - cold initialization where initial data are read from a METIS simulation file.
  - hot initialization where the initial data are computed by running METIS.

- metis_computation
  METIS will compute new data at the time prescribed by the actor using previously known plasma state. This time can be:
  - in the future to compute the evolution of the plasma state,
  - the same time as the previous one to perform a convergence loop in the workflow
  - in the past to restart from a previous converged state of the workflow in the case of non-convergence in some actor.

The clock in this actor can be provided either by:

1. another actor, on:
   - pulse schedule port
   - equilibrium port
   - plasma_profiles port
   - core_profiles port
   - plasma_sources port
   - core_sources port
   - clock port

   in this order of priority; using either time_stamp or next_time_stamp depending on the settings.
2. METIS simulation file as prescribed in the settings
3. Internal actor clock parametrized in the settings

The actor has 3 sets of input ports:

1. one set for initialization with operator F_INIT. These input ports are read at the initialization of the METIS actor.
2. one set with operator F_INIT read at each subsequent call (not at initialization)
3. one set with operator S read at each subsequent call (not at initialization)

Only the connected ports are read. If both F_INIT and S ports are connected, the data from the S port overwrite the data from the F_INIT port.

Input ports are:

- clock
  Uses only the time_stamp or next_time_stamp information from the message, can be any IDS. This port can be used to impose an external clock to the actor when METIS actor does

not use any IDS data in input. This port can be connected to any port of any other actor to synchronize METIS internal clock on this actor clock.

- pulse_schedule
  contains the main input data for METIS.

- equilibrium
  contains data for external equilibrium and/or external current diffusion.

- plasma_profiles or core_profiles
  kinetic profiles used for external data as electron temperature or density, …

- plasma_source or core_sources
  data for external sources (heat and/or current).

The name of the port and the name of the IDS sent in the message are identical.

The IDS subset used by the METIS actor from each IDS is described in the document named "METIS_inputs_from_MAS_IDSs.pdf" in the directory "doc" under the root directory of METIS installation.

The METIS MUSCLE 3 actor output two sets of ports. The first one uses the operator O_F and the second one uses the operator O_I. Please refer to the MUSCLE 3 documentation for detailed information on the use of these operators. Both sets of ports can be used simultaneously. Only the IDS data of the connected port are mapped. Each set of ports has the following ports:

- pulse_schedule
- equilibrium
- plasma_profiles
- core_profiles
- plasma_sources
- core_sources
- dataset_description
- dataset_fair
- summary
- plasma_transport
- core_transport
- radiation
- edge
- transport_solver_mumerics (obsolete in DD v4.x)

Each port sends in its message the serialized data of the IDS of same name with the exception of edge port. The edge port sends in it message the serialized data the two IDSs edge_profiles and edge_transport. These two IDSs are mapped only when the 2-point model is switched on in METIS.

Fields filled in each IDS are described in the document "metis_imas_output_documentation.md" under the root directory of METIS installation (this document is dynamically generated and can be updated by the user from a METIS IMAS data set).

Remark 1: As METIS actor has a set of input ports and set of output ports for each MUSCLE 3 operator (F_INIT, S , O_I and O_F) and an additional set of input ports for the initialization of the actor, METIS actor is agnostic of the workflow content and is sufficiently versatile to cover all known use cases.

Remark 2: the settings of the actor are described in a next section.

Organization of the code for the actor

The code of the actor is stored under METIS root directory in the subdirectory “workflow/muscle3”. In this directory the code has the following organization:

1. mfile: this directory contains the main MATLAB sources for the actor.
2. YMMSL: this directory contains some examples for the configuration of the actor (extended YAML files).
3. run_script: this directory contains some examples of shell script used to launch examples.

In the mfile subdirectory, the main actor file is “metis4muscle3.m”. The input and output ports are declared in “metis4muscle3_port_list.m”, the actor settings in “metis4muscle3_setting_list.m” and the external data in metis4muscle3_listofexternal_data.m. The METIS internal parameters are declared in the file “<METIS root directory>/zerod/zerod_param.m”. The default values of each of these parameters can be read in the corresponding declaration file. Other files have explicit name for their functions.

Call of METIS actor and input parameters

The MATLAB actor is launched by the workflow manager of MUSCLE 3 using the configuration YYMSL file. The main MATLAB function for the actor is “metis4msucle3.m”. The syntax is:

```
metis4muscle3(debug_level,checkpoint_on_off)
```

The function has two inputs arguments:

1. debug_level
   Level of debug and MATLAB profiler state. If “debug_level” is negative, the MATLAB profiler is switched on and a report is created at the end of the simulation in the workdir of the METIS instance. The possible values of abs(debug_level) are:
   0 = no debug mechanism
   1 = MATLAB logfile is written in workdir of the METIS instance (debug_metis_in_muscle3.txt)

   2 = same as option 1 and debugger set to stop on error

   3 = same as option 2 and a keyboard gives the hand on interactive debug at the start of the function (keyboard insertion, work only in interactive mode of MATLAB).

   Additionally, if debug_level is non zero, the internal variables of the function metis4muscle3 are saved in the workdir of the METIS (meti4muscle3_caller_*_context.mat) and the MATLAB workspace is also saved in the workdir of the METIS instance (meti4muscle3_workspace_context.mat).

2. checkpoint_on_off
   if true, turns on the checkpointing features of the actor and allows creation of checkpoint dataset and cold or hot restart of the workflow from a checkpoint dataset (see https://muscle3.readthedocs.io/en/latest/checkpointing.html for details information).

Remark 1:

The actor metis4muscle3 can be launched in an interactive MATLAB (option -r of MATLAB command) or in batch mode (option -batch of MATLAB command). The interactive mode is recommended for debugging and the batch mode is recommended for standard execution as the start of MATLAB is faster. For more details, see the MATLAB documentation for the execution of the "matlab" code (https://www.mathworks.com/help/matlab/ref/matlablinux.html?searchHighlight=linux+command&s_tid=srchtitle_support_results_3_linux+command).

Remark 2:

MATLAB is a multithreaded code and METIS can take advantage of the feature. It is possible to speed-up the execution of the METIS actor by allocating many cores to the execution of the actor in the YMMSL configuration file (in resource section of METIS actor instance, the parameter "thread"). MATLAB can use up to 64 cores and METIS has shown efficient speed-up at least up to 30 cores.

Remark 3:

The command line for the call of "metis4muscle3" is part of YYMSL file. The default syntax is:

```
implementations:
  metis:
    executable: matlab
    args: {-r, -batch}  "[s,t] = unix('which python');
                         pyenv('Version',strtrim(t),'ExecutionMode','InProcess');
                         addpath(getenv('DIR_METIS4MUSCLE3'));
                          metis4muscle3(<debug_level>, <checkpoint_on_off>);"
```

Depending of the use, Matlab should be launch either with the argument "-r" or "-batch" (see above).

The two first commands set the correct Python engine used from Matlab. The environment variable "DIR_METIS4MUSCLE3" should be set to point to the directory where the mfile "metis4muscle3.m" is located.

Files written in the "workdir" of METIS instance

During the execution of the METIS MUSCLE 3 actor some files are written in the workdir (<run MUSCLE 3 directory>/instances/<metis instance name>/workdir):

- at any debug_level value:
  METIS_actor_for_MUSCLE3_layout.png → dynamically generated layout of METIS MUSCLE 3 actor (figure 1 is an example of such image).

- with debug switch on:
  debug_metis_in_muscle3.txt → details MATLAB logfile
  meti4muscle3_caller_finish_context.mat → MATLAB caller workspace (mainly internal variables of METIS MUSCLE3 actor)
  meti4muscle3_workspace_context.mat → MATLAB workspace (mainly variables of MATLAB workspace)

Example and test case

An example of YMMSL file with a quite complete set of settings is "metis4muscle3_full_test.ymmsl" in the directory "worflow/muscle3/YMMSL" under METIS root directory. The shell script for running a test case with this configuration file is "launch_metis4muscle3_full_test.bash" in in the directory " workflow/muscle3/run_script" under METIS root directory. The content of this shell script should be updated for the local environment where it will be launched. This shell script example will launch three Matlab actors:

- scheduler → this actor generates IDSs from METIS and streams their content.
- metis → MUSCLE 3 actor for METIS
- listener → this actor receives messages containing IDSs from METIS and displays messages sizes.

METIS actor settings

Introduction

METIS MUSCLE 3 actor has a large set of settings. Each setting has a default value. There are no mandatory settings that must be declared in the YMMSL file. Actor settings are split in three categories:

1. Actor configuration parameters that are specific to the actor
2. METIS internal parameters (see metis_documentation_for_parameters_expert_mode.pdf in "doc" directory under METIS root directory for details information).
3. Switches controlling the use of external data in METIS (see METIS_inputs_from_IMAS_IDSs.pdf in "doc" directory under METIS root directory and external_data_rule_for_METIS.m in "zerod" directory under METIS root directory for details information).

In the following, the default value of each parameter is noted in brackets.

Actor configuration

Workflow synchronisation

- metis_synchronization_reuse (0)
  synchronization on "reuse_instance()" method: METIS actor can be forced to be synchronized or not on "reuse_instance" state.
  Warning: the use of this option can freeze the workflow.

- metis_synchronization_flush (0)
  Force buffers to be flushed when O_I ports are used to prevent workflow dead lock or message synchronization problem on some Linux systems with MUSCLE3.

- metis_alwaysreadinitport (0)
  if = 1, the actor will read also initialization ports on every iteration in addition of other input ports and not only at initialization. The use of this option is not recommended.

- metis_updatetimeonrestart (0)
  if =1, the actor will update its internal clock on restart with resuming message time stamp value.

Workflow clock

- metis_clock_time_start (0.0)
  time at which start the simulation in replay mode or starting time for the internal clock (s). In replay mode, if this time is outside the simulated times, the replay will start from the first time slice of the simulated data.

- metis_clock_time_stop (Inf)
  time at which the simulation will end (In replay mode, the simulation stop at the minimum of this time and the end of simulated data. Can be set to large number to be switched off).

- metis_clock_time_dt (3.14)
  time step (s): used if no external clock (from another actor) is provided to the actor or if internal clock is used.

- metis_clock_internal (off)
  switch “on” or “off” internal actor clock.

- metis_usenexttimestamp (0)
  if = 1, METIS actor will compute data evolution for “next_timestamp” instead of “timestamp” (see https://muscle3.readthedocs.io/en/latest/python_api.html#libmuscle.Message.next_timestamp for details).

METIS initialisation and behaviour

- metis_filename (empty string)
  METIS file name (with full path) used for replay mode or to provide input data to METIS if no IDS “pulse_schedule” is provided. It is also used in cold initialization mode (see setting “metis_coldinit”).

- metis_computation (1)
  METIS actor computation can be switched off (0) and data are interpolated on time from METIS simulation file defined by setting “metis_filename”. Otherwise, METIS is computed at

each actor call.

- metis_workingpoint (0)
  if = 1, METIS is run in working point mode (mainly for power plant design). In this case METIS solve a non-timed problem and search for the steady state solution. The clock of the workflow has then no impact on the solution.

- metis_coldinit (0)
  allows to switch on cold initialization of METIS: if =1, METIS is initialized from METIS simulation file defined by setting "metis_filename" instead of performing a first computation run. The data are time interpolated to match the workflow clock.

External data steering (see next section for selection of external data used in the workflow)

- metis_idstime (0)
  if = 1, external data will be timed on time stored in IDS instead of time stamp from the corresponding MUSCLE3 message.

- metis_timeoffset (0.0)
  Offset between METIS internal time (time stamp in METIS output message) and input message time stamp or IDS time used for external data (s).

- metis_nbpts1d (101)
  Number of radial points in 1D profile used for external data at the exception of the data for external equilibrium which has a fixed internal number of radial points. This value can be reduced up to 21 at the price of reduced precision in the mapping of the radial coordinate.

- metis_psioffset (0.0)
  Poloidal flux offset between METIS internal value and external equilibrium value:
  psi_metis = cocos_sign * psi_equilibrium / (2*pi) – metis_psioffset (Wb/rad).

- metis_numberofequilibriumtimeslices (3)
  Number of external equilibrium time slices stored in METIS actor. The minimum is 3. Should be set to Inf if the workflow is intended to rewind back in time for convergence reasons. It has an impact on the memory used by the actor.

Tuning of sawtooth mimicking mechanism (to be used with NICE code in evolutive mode)

- metis_sawtooth_onoff (off)
  Turn on or off mechanism for mimicking sawtooth when METIS is used in link with the NICE code solving Grad-Shafranov equation and diamagnetic function evolution (equivalent to current diffusion equation). The sawtooth behavior is configured by using METIS internal parameters. This mechanism can be useful to prevent too low central safety factor and too high internal inductance increases during coupled simulation.

- metis_sawtooth_nbtmin (3)
  minimum number of time steps between two sawtooth trigger events.

METIS parameter

Each METIS internal parameter can be overwritten by a parameter declared in the section setting of the YMMSL file. Usually, these parameters are set either in the "pulse schedule" IDS code parameters (for details, see METIS_inputs_from_IMAS_IDSs.pdf in "doc" directory under METIS root directory) or in the METIS simulation file (see metis_documentation_for_parameters_expert_mode.pdf in "doc" directory under METIS root directory for details information). If the parameter is declared in the setting of the actor in the YMMSL file, that value will overwrite the initial METIS settings. The name of the parameter should be formatted as "metis_option_<METIS parameter name or label>" (see file zerod_param.m in "zerod" directory under METIS root directory for details information about parameter name or label). The "prefix metis_option_" is mandatory. The range or possible values and default value of these parameters are also described in zerod_param.m and the detailed description of these parameters is available in metis_documentation_for_parameters_expert_mode.pdf.

You can see hereafter some examples of such parameters:

1. the machine name can be updated to the running simulation:
    - metis_option_machine: test_muscle3

2. When METIS is used with NICE code in inverse mode, the available flux can be updated for the current simulation (this parameter changes also the pre-magnetization state for the initial inverse equilibrium computed by NICE):
    - metis_option_available_flux: 130.0

3. Toroidal magnetic field and current plasma orientations can be wrong in the initial simulation as they have nearly no impact on METIS results and can be updated before doing a simulation embedding other IMAS actors:
    - metis_option_sign: 1
    - metis_option_orientation: -1

4. If the simulation has been done for another version of IMAS, the declared COCOS can be wrong and some check can be useful (note that COCOS check takes some computing time and can slow down the simulation):
    - metis_option_COCOS_method: Sauter
    - metis_option_COCOS_check: on

External data switches

The use in METIS of external data can be controlled using specific settings. The use of each external data is effective only if the corresponding port is connected and the message read on this port contains a non-empty IDS (see METIS_inputs_from_IMAS_IDSs.pdf in "doc" directory under METIS root directory and external_data_rule_for_METIS.m in zerod directory under METIS root directory for details information).

The value of the parameter is 0 to switch off the use of the IDS data as an external data in METIS and is 1 to switch on the use of the IDS data as an external data in METIS.

From equilibrium IDS

- metis_external_data_equilibrium (0)
  Control the use of equilibrium IDS data as external data equilibrium in METIS

- metis_external_data_current_diffusion (0)
  Control the use of timed equilibrium IDS data as external data for current diffusion in METIS. This is switched off at the initialization of the actor as time derivative quantities required for the computation are not available. It is also ineffective in working point mode as METIS converges to the steady state of the plasma. When this feature is used, it is recommended to also switch on the use of an external equilibrium (metis_external_data_equilibrium=1).

From IDS plasma_profiles or core_profiles

Remark: these data require also the equilibrium IDS for precise coordinates transformation.

- metis_external_data_electron_temperature (0)
  Control the use of plasma_profiles or core_profiles IDS data as external data for electron temperature (T_e) in METIS.

- metis_external_data_electron_density (0)
  Control the use of plasma_profiles or core_profiles IDS data as external data for electron density shape (~ n_e) in METIS.

- metis_external_data_ion_temperature (0)
  Control the use of plasma_profiles or core_profiles IDS data as external data for ion temperature (T_i) in METIS.

- metis_external_data_charge_effective (0)
  Control the use of plasma_profiles or core_profiles IDS data as external data for effective charge shape (~ Zeff) in METIS.

- metis_external_data_momentum (0)
  Control the use of plasma_profiles or core_profiles IDS data as external data for plasma toroidal rotation (vtor) in METIS.

From IDS plasma_sources

Remark: these data require also equilibrium IDS for precise coordinates transformation.

- metis_external_data_LHCD (0)
  Control the use of plasma_sources or core_sources IDS data as external data for source shape (current and heat) for LHCD in METIS.

- metis_external_data_ECRH (0)
  Control the use of plasma_sources or core_sources IDS data as external data for source shape (current and heat) for ECRH or ECCD in METIS.

- metis_external_data_ICRH (0)
  Control the use of plasma_sources or core_sources IDS data as external data for source shape (current and heat) for ICRH, FWEH or FWCD in METIS.

- metis_external_data_NBI (0)
  Control the use of plasma_sources or core_sources IDS data as external data for source shape (current and heat) for NBI or NBICD in METIS.

- metis_external_data_PLINE (0)
  Control the use of plasma_sources or core_sources IDS data as external data for line radiative power density well in METIS.

- metis_external_data_runaway (0)
  Control the use of plasma_sources or core_sources IDS data as external data for runaway electrons current density source in METIS (runaway electrons are seen here as a source in current diffusion equation).

F.A.Q

Not yet!